Supplementary Material of

Regime-Dependent Bitcoin Volatility Forecasting under Alternative Information Sets: Evidence from Econometric, Deep Learning, and Hybrid Models

Descriptive Statistics

Table S1 presents descriptive statistics for all variables in the merged dataset, which was originally compiled for the period from 1 February 2018 to 1 February 2026. The table reports the number of observations, mean, median, standard deviation, minimum, maximum, skewness, kurtosis, and the Jarque–Bera test statistic for normality. Since the sentiment and blockchain variables are drawn from diverse data sources with non-overlapping time spans and occasional missing entries, the sample size is not identical across series. Consequently, the descriptive statistics for each variable are calculated using all observations available for that specific series over the entire collection period.

For the purposes of model estimation and forecast assessment, the empirical analysis relies on a balanced subsample that includes only those observations for which the volatility proxy and all chosen explanatory variables are simultaneously available. Consequently, the final estimation and evaluation sample employed in the main paper covers the period from 1 February 2018 to 19 July 2025. Limiting the forecasting exercise to this balanced time span guarantees comparability across models and information sets and prevents distortions that might stem from sparse or missing observations in the later portion of the collected dataset.

Table S1. Descriptive Statistics of All Variables

Variable	N	Mean	Median	Std. Dev.	Min	Max	Skewness	Kurtosis	JB Stat.	JB p-value
Opening Price (USD)	2923	38,086.6100	28,038.9700	32,572.1400	3,236.2750	124,752.1000	0.9313	2.7663	429.1927	0.0000
Highest Price (USD)	2923	38,831.0200	28,502.6900	33,094.5800	3,275.3780	126,198.1000	0.9169	2.7305	418.3971	0.0000
Lowest Price (USD)	2923	37,294.7400	27,400.3100	32,006.3100	3,191.3030	123,196.0000	0.9470	2.8065	441.4272	0.0000
Closing Price (USD)	2923	38,108.5500	28,044.1400	32,575.9500	3,236.7620	124,752.5000	0.9298	2.7631	428.0063	0.0000
Logarithmic Return	2923	0.0007	0.0007	0.0337	-0.4647	0.1718	-0.9501	18.3708	29,214.4605	0.0000
Trading Volume (USD)	2923	1.5771e+15	6.5595e+14	2.2012e+15	1.1496e+13	2.1782e+16	2.4923	11.2527	11,320.9184	0.0000
Volatility proxy (forecast target)	2722	3.1519	2.9449	1.6133	0.5945	15.9863	3.5728	26.8865	70,502.9342	0.0000
Close-to-Close Volatility	2923	3.0381	2.7847	1.3812	0.5820	10.9076	1.5597	7.6598	3,829.6634	0.0000
High–Low Range (%)	2923	4.2342	3.4467	3.2987	0.3587	61.7359	3.8357	42.1928	194,248.7270	0.0000
Average True Range (ATR)	2923	1,531.3410	1,046.7240	1,304.8940	57.9237	6,057.9470	0.8223	2.7293	338.3649	0.0000
Relative Strength Index (14-day)	2729	47.3450	47.9362	17.9047	0.6629	95.2156	-0.0584	2.5498	24.6048	0.0000
Bollinger Band Width	2923	0.2149	0.1767	0.1385	0.0267	0.9245	1.5808	6.1442	2,421.4974	0.0000
News Sentiment Tone	2742	0.0432	0.0769	2.3871	-6.9086	10.0000	0.0426	2.7805	6.3376	0.0421
Twitter Sentiment Score	2742	0.0012	0.0041	0.3106	-1.0000	1.0000	-0.0402	3.0072	0.7459	0.6887
Fear–Greed Index	2919	46.9784	47.0000	21.6846	5.0000	95.0000	0.1736	1.9696	143.7960	0.0000
Transaction Count (daily)	2922	327,742.7655	301,770.7500	118,840.8027	124,640.0000	859,629.0000	1.4205	5.1779	1,560.1919	0.0000

Transaction Count (7-day MA)	2742	336,279.8000	308,222.2000	116,450.7000	124,640.0000	859,629.0000	1.4280	5.0834	1,427.8456	0.0000
Transaction Volume (USD)	2742	3.8125e+09	2.5558e+09	3.5720e+09	2.2351e+08	2.8380e+10	1.9055	8.0143	4,531.9684	0.0000
Network Hashrate (EH/s)	2742	2.8249e+08	1.6669e+08	2.5830e+08	1.9292e+07	1.0389e+09	1.1125	3.0512	565.8846	0.0000

Notes: *Descriptive statistics are computed using all available observations for each variable over the full collected sample period from 1 February 2018 to 1 February 2026. Due to staggered availability and occasional missing values in sentiment and blockchain series, the number of observations differs across variables. The final balanced sample used for model estimation and forecast evaluation in the main analysis spans from 1 February 2018 to 19 July 2025. The Jarque–Bera statistic tests the null hypothesis of normality.*

Across the full sample period, Bitcoin prices display marked dispersion, reflecting distinct boom–bust cycles and considerable long-term fluctuations in market value. The distribution of price levels is moderately skewed to the right and exhibits excess kurtosis, indicating asymmetric upside movements coupled with sharp corrections. Average log returns are close to zero, yet they show substantial volatility, negative skewness, and strong leptokurtosis. The Jarque–Bera test strongly rejects the null of normality, confirming fat tails and frequent extreme realizations, consistent with well-established stylized facts for financial—and especially cryptocurrency—return series (Cont, 2001). The volatility proxy used as the forecasting regressor is similarly heavily right-skewed and highly leptokurtic, reflecting frequent volatility spikes and persistent clustering. Range-based measures, including close-to-close volatility and the high–low percentage range, reveal comparable distributional characteristics. Taken together, these properties signal conditional heteroskedasticity and regime-varying variance dynamics, thereby motivating the use of ARCH/GARCH-type and other nonlinear forecasting frameworks (Engle, 1982; Bollerslev, 1986).

Technical indicators exhibit bounded and distinctly nonlinear distributions. The Relative Strength Index (RSI) is naturally restricted to a fixed interval and displays minimal skewness, whereas Bollinger Band width is strongly right-skewed, reflecting sharp expansions of the bands during high-volatility episodes. These features suggest that technical indicators predominantly capture short-term trading dynamics and volatility states, rather than resembling Gaussian noise. Sentiment metrics are also non-normal, though generally less extreme than return and volatility distributions. Sentiment extracted from news and social media clusters around neutral levels yet still diverges from normality, while the Fear–Greed Index is range-bounded and only slightly asymmetric. Jarque–Bera tests reject normality for most sentiment measures, consistent with behavioral finance evidence that investor sentiment affects asset prices in nonlinear, regime-dependent ways (Baker and Wurgler, 2007; Tetlock, 2007). Twitter-based sentiment is the key exception, showing a relatively symmetric, bounded distribution that does not lead to rejection of normality.

Blockchain activity metrics exhibit pronounced right skewness and substantial dispersion. Daily transaction counts and their seven-day moving averages display strong positive skew and excess kurtosis, reflecting both steady network growth and intermittent surges in usage. USD-denominated transaction volume is even more dispersed, capturing sharp jumps in capital flows and periods of intensified trading. The network hash rate shows comparable volatility and a heavily right-skewed distribution, consistent with long-run expansion and occasional disruptions in mining activity. Differences in the number of observations between blockchain variables and sentiment series stem from varying data source coverage and do not meaningfully affect the interpretation of their marginal distributions. Overall, the dataset displays heavy tails, volatility clustering, asymmetry, and structural growth—features that are standard in cryptocurrency markets. The widespread rejection of normality for returns, volatility, and several covariates further supports the use of conditional variance models and flexible nonlinear forecasting approaches capable of handling extreme outcomes and regime shifts.

Additional Supplementary Forecast-Evaluation Tables

The following supplementary tables report additional empirical outputs from the corrected rolling out-of-sample forecasting exercise. They are intended to document the full model rankings and Diebold–Mariano test evidence underlying the condensed results reported in the main manuscript. In particular, while the main paper reports the best-performing family representatives, the supplementary tables provide the complete normal-period and crisis-period rankings across the 27 model specifications, together with statistical comparisons against the relevant period- or regime-best benchmark.

Table S2 reports the full normal-period ranking of all 27 model specifications. Unlike Table 4 in the main manuscript, which presents only the best-performing model from each model family, this table reports the complete QLIKE-based ranking across traditional volatility models, standalone LSTMs, and state-augmented hybrid GARCH–LSTM specifications. The Diebold–Mariano test compares each model with the normal-period benchmark, LSTM-T, using QLIKE loss differentials.

Table S2. Full Normal-Period Ranking of All 27 Model Specifications

Model family	Model	Observations	QLIKE	RMSE	MAE	Overall rank	Best model in regime	DM vs best	p-value vs best	Conclusion vs best
LSTM	LSTM-T	1583	0.1934	0.8425	0.6608	1	LSTM-T			Best benchmark
Hybrid	Hybrid_GARCH_T	1583	0.2062	0.8925	0.6984	2	LSTM-T	0.7941	0.4273	Not significantly different from best
Hybrid	Hybrid_APARCH_T	1583	0.2078	0.9096	0.6947	3	LSTM-T	0.8182	0.4134	Not significantly different from best
Hybrid	Hybrid_iGARCH_T	1583	0.2128	0.9460	0.7230	4	LSTM-T	1.0554	0.2914	Not significantly different from best
Hybrid	Hybrid_EGARCH_T	1583	0.2133	0.9190	0.7030	5	LSTM-T	0.8852	0.3762	Not significantly different from best
Hybrid	Hybrid_EGARCH_TSB	1583	0.2149	0.9501	0.7095	6	LSTM-T	0.9801	0.3272	Not significantly different from best
Hybrid	Hybrid_TGARCH_TS	1583	0.2189	1.0311	0.7562	7	LSTM-T	0.9342	0.3503	Not significantly different from best
Hybrid	Hybrid_GARCH_TS	1583	0.2207	1.0895	0.7904	8	LSTM-T	1.0711	0.2843	Not significantly different from best
Traditional	TGARCH	1583	0.2209	1.0038	0.8047	9	LSTM-T	1.0520	0.2930	Not significantly different from best
Traditional	GARCH	1583	0.2218	1.0064	0.8106	10	LSTM-T	1.1011	0.2710	Not significantly different from best
Hybrid	Hybrid_GJR_GARCH_T	1583	0.2222	0.9359	0.7080	11	LSTM-T	1.6033	0.1091	Not significantly different from best
Hybrid	Hybrid_iGARCH_TS	1583	0.2238	1.1534	0.7937	12	LSTM-T	1.2788	0.2012	Not significantly different from best
Hybrid	Hybrid_APARCH_TS	1583	0.2248	1.1590	0.8145	13	LSTM-T	1.2052	0.2283	Not significantly different from best
Traditional	GJR-GARCH	1583	0.2292	1.0491	0.8286	14	LSTM-T	1.3685	0.1713	Not significantly different from best
LSTM	LSTM-TS	1583	0.2335	1.0208	0.7727	15	LSTM-T	1.6367	0.1019	Not significantly different from best
Hybrid	Hybrid_EGARCH_TS	1583	0.2357	1.2311	0.8257	16	LSTM-T	1.4674	0.1425	Not significantly different from best
Hybrid	Hybrid_GJR_GARCH_TSB	1583	0.2376	1.0443	0.7578	17	LSTM-T	1.5353	0.1249	Not significantly different from best
Traditional	iGARCH	1583	0.2410	1.1596	0.9116	18	LSTM-T	1.8238	0.0684	Marginally worse than best at 10%
LSTM	LSTM-TSB	1583	0.2472	1.2775	0.8434	19	LSTM-T	1.9466	0.0518	Marginally worse than best at 10%
Hybrid	Hybrid_GJR_GARCH_TS	1583	0.2589	1.1540	0.7802	20	LSTM-T	1.4072	0.1596	Not significantly different from best
Hybrid	Hybrid_APARCH_TSB	1583	0.2615	1.1313	0.7799	21	LSTM-T	2.1289	0.0334	Significantly worse than best at 5%
Hybrid	Hybrid_TGARCH_T	1583	0.2631	1.0366	0.7746	22	LSTM-T	2.8883	0.0039	Significantly worse than best at 1%
Hybrid	Hybrid_iGARCH_TSB	1583	0.2677	1.2657	0.8466	23	LSTM-T	2.6081	0.0092	Significantly worse than best at 1%
Hybrid	Hybrid_GARCH_TSB	1583	0.2761	1.3147	0.8486	24	LSTM-T	2.9982	0.0028	Significantly worse than best at 1%
Hybrid	Hybrid_TGARCH_TSB	1583	0.2799	1.2867	0.8524	25	LSTM-T	3.1639	0.0016	Significantly worse than best at 1%
Traditional	APARCH	1583	0.5465	2.2598E+30	5.6798E+28	26	LSTM-T	2.8277	0.0047	Significantly worse than best at 1%
Traditional	EGARCH	1583	1.4943E+10	1.6868	0.8265	27	LSTM-T	1.0035	0.3158	Not significantly different from best

Notes: *Models are ranked by QLIKE, where lower values indicate better volatility-forecast accuracy. The normal-period benchmark is LSTM-T. The DM statistic is computed from QLIKE loss differentials relative to LSTM-T. A p-value above 0.10 indicates that the model is not statistically worse than the benchmark at conventional levels. RMSE and*

MAE are reported as complementary scale-based diagnostics. Very large QLIKE, RMSE, or MAE values indicate numerical instability and should be interpreted cautiously even when the DM test does not reject equal predictive accuracy.

Table S3 extends the crisis-period analysis by reporting the full regime-specific ranking of all 27 model specifications. For each crisis window, all traditional, LSTM, and hybrid models are re-evaluated directly within that regime rather than carrying forward the normal-period winners. This table therefore provides the detailed evidence behind Table 5 in the main manuscript and shows whether crisis-period forecast gains arise from model architecture, information set, or regime characteristics.

Table S3. Full Crisis-Period Ranking of All 27 Model Specifications by Regime

Regime	Model family	Model	Observations	QLIKE	RMSE	MAE	Overall rank	Best model in regime	DM vs best	p-value vs best	Conclusion vs best
COVID-19 Shock	Hybrid	Hybrid_TGARCH_TS	102	1.2636	3.9421	2.2698	1	Hybrid_TGARCH_TS			Best benchmark
	Hybrid	Hybrid_GARCH_TSB	102	1.2973	4.0876	2.5545	2	Hybrid_TGARCH_TS	0.0538	0.9572	Not significantly different from best
	Hybrid	Hybrid_GJR_GARCH_TS	102	1.3781	3.9549	2.2367	3	Hybrid_TGARCH_TS	0.2823	0.7783	Not significantly different from best
	Hybrid	Hybrid_EGARCH_TS	102	1.4659	4.2148	2.4783	4	Hybrid_TGARCH_TS	0.8059	0.4222	Not significantly different from best
	Hybrid	Hybrid_APARCH_TS	102	1.5531	4.4382	2.6217	5	Hybrid_TGARCH_TS	0.6017	0.5487	Not significantly different from best
	Hybrid	Hybrid_GJR_GARCH_TSB	102	1.9372	4.8811	2.7498	6	Hybrid_TGARCH_TS	1.2078	0.2299	Not significantly different from best
	Hybrid	Hybrid_iGARCH_TSB	102	2.0066	4.6964	2.8062	7	Hybrid_TGARCH_TS	1.2787	0.2039	Not significantly different from best
	Hybrid	Hybrid_GARCH_TS	102	2.1424	4.8499	2.8933	8	Hybrid_TGARCH_TS	1.3628	0.1760	Not significantly different from best
	LSTM	LSTM-TSB	102	2.2881	4.5884	2.7851	9	Hybrid_TGARCH_TS	1.0447	0.2987	Not significantly different from best
	Hybrid	Hybrid_iGARCH_TS	102	2.5704	5.1526	2.9909	10	Hybrid_TGARCH_TS	1.9639	0.0523	Marginally worse than best at 10%
	LSTM	LSTM-TS	102	3.3390	5.1514	2.8273	11	Hybrid_TGARCH_TS	2.1026	0.0380	Significantly worse than best at 5%
	Hybrid	Hybrid_APARCH_TSB	102	3.3515	4.8084	3.1635	12	Hybrid_TGARCH_TS	2.1379	0.0349	Significantly worse than best at 5%
	Traditional	iGARCH	102	3.3733	6.4692	4.2240	13	Hybrid_TGARCH_TS	1.8257	0.0708	Marginally worse than best at 10%
	Hybrid	Hybrid_EGARCH_T	102	3.5917	5.5277	3.6828	14	Hybrid_TGARCH_TS	2.0292	0.0451	Significantly worse than best at 5%
	Traditional	EGARCH	102	3.7178	6.5996	3.7986	15	Hybrid_TGARCH_TS	1.9297	0.0565	Marginally worse than best at 10%
	Traditional	TGARCH	102	3.7804	5.6498	3.3840	16	Hybrid_TGARCH_TS	1.9206	0.0576	Marginally worse than best at 10%
	Traditional	APARCH	102	3.8307	15.8896	5.9682	17	Hybrid_TGARCH_TS	2.0737	0.0407	Significantly worse than best at 5%
	LSTM	LSTM-T	102	3.9892	5.4100	3.2455	18	Hybrid_TGARCH_TS	1.8926	0.0613	Marginally worse than best at 10%
	Traditional	GJR-GARCH	102	3.9939	6.7049	4.0985	19	Hybrid_TGARCH_TS	2.0206	0.0460	Significantly worse than best at 5%
	Traditional	GARCH	102	4.0078	6.1953	3.8705	20	Hybrid_TGARCH_TS	2.0129	0.0468	Significantly worse than best at 5%
	Hybrid	Hybrid_iGARCH_T	102	4.1713	5.4965	3.2517	21	Hybrid_TGARCH_TS	2.0231	0.0457	Significantly worse than best at 5%
	Hybrid	Hybrid_APARCH_T	102	4.3988	5.5316	3.4737	22	Hybrid_TGARCH_TS	2.0012	0.0480	Significantly worse than best at 5%
	Hybrid	Hybrid_GARCH_T	102	4.5377	5.5184	3.1021	23	Hybrid_TGARCH_TS	2.1274	0.0358	Significantly worse than best at 5%
	Hybrid	Hybrid_GJR_GARCH_T	102	4.5573	5.4645	2.9637	24	Hybrid_TGARCH_TS	2.0415	0.0438	Significantly worse than best at 5%
	Hybrid	Hybrid_TGARCH_T	102	4.7575	5.5226	3.1575	25	Hybrid_TGARCH_TS	2.0786	0.0402	Significantly worse than best at 5%
	Hybrid	Hybrid_TGARCH_TSB	102	25.4308	5.1059	3.1975	26	Hybrid_TGARCH_TS	1.2757	0.2050	Not significantly different from best
	Hybrid	Hybrid_EGARCH_TSB	102	2.0351E+11	5.0151	3.0696	27	Hybrid_TGARCH_TS	1.0263	0.3072	Not significantly different from best
China Mining Ban	Hybrid	Hybrid_EGARCH_TSB	92	0.1528	1.1910	0.9108	1	Hybrid_EGARCH_TSB			Best benchmark
	Traditional	GARCH	92	0.1870	1.5488	1.2220	2	Hybrid_EGARCH_TSB	0.7732	0.4414	Not significantly different from best
	Traditional	EGARCH	92	0.1908	1.5052	1.1961	3	Hybrid_EGARCH_TSB	1.0170	0.3119	Not significantly different from best
	Traditional	iGARCH	92	0.2002	1.7170	1.3436	4	Hybrid_EGARCH_TSB	0.8912	0.3752	Not significantly different from best

	Traditional	TGARCH	92	0.2013	1.6229	1.2861	5	Hybrid_EGARCH_TSB	1.1375	0.2583	Not significantly different from best
	Hybrid	Hybrid_EGARCH_TS	92	0.2055	1.4003	1.1167	6	Hybrid_EGARCH_TSB	1.4880	0.1402	Not significantly different from best
	Traditional	GJR-GARCH	92	0.2438	1.9080	1.4113	7	Hybrid_EGARCH_TSB	1.9751	0.0513	Marginally worse than best at 10%
	Hybrid	Hybrid_GJR_GARCH_TS	92	0.2494	1.4130	1.0686	8	Hybrid_EGARCH_TSB	1.5358	0.1281	Not significantly different from best
	Hybrid	Hybrid_APARCH_TS	92	0.2657	1.4776	1.1587	9	Hybrid_EGARCH_TSB	1.5912	0.1150	Not significantly different from best
	Hybrid	Hybrid_TGARCH_TS	92	0.2883	1.4209	1.0775	10	Hybrid_EGARCH_TSB	1.1725	0.2441	Not significantly different from best
	Traditional	APARCH	92	0.3019	2.1271	1.5127	11	Hybrid_EGARCH_TSB	2.8551	0.0053	Significantly worse than best at 1%
	Hybrid	Hybrid_iGARCH_TSB	92	0.3232	1.4806	1.0875	12	Hybrid_EGARCH_TSB	1.8851	0.0626	Marginally worse than best at 10%
	Hybrid	Hybrid_GARCH_TSB	92	0.3268	1.5965	1.1876	13	Hybrid_EGARCH_TSB	2.4522	0.0161	Significantly worse than best at 5%
	Hybrid	Hybrid_iGARCH_TS	92	0.3530	1.5383	1.1898	14	Hybrid_EGARCH_TSB	1.4078	0.1626	Not significantly different from best
	Hybrid	Hybrid_GARCH_TS	92	0.3633	1.7783	1.1798	15	Hybrid_EGARCH_TSB	1.5722	0.1194	Not significantly different from best
	Hybrid	Hybrid_APARCH_TSB	92	0.3648	1.5999	1.1467	16	Hybrid_EGARCH_TSB	1.9995	0.0485	Significantly worse than best at 5%
	Hybrid	Hybrid_GARCH_T	92	0.3808	1.5510	1.1662	17	Hybrid_EGARCH_TSB	1.9397	0.0555	Marginally worse than best at 10%
	LSTM	LSTM-T	92	0.4074	1.5582	1.1785	18	Hybrid_EGARCH_TSB	1.5633	0.1215	Not significantly different from best
	Hybrid	Hybrid_APARCH_T	92	0.4273	1.6797	1.3160	19	Hybrid_EGARCH_TSB	2.3079	0.0233	Significantly worse than best at 5%
	Hybrid	Hybrid_TGARCH_TSB	92	0.4380	1.7114	1.2725	20	Hybrid_EGARCH_TSB	2.6517	0.0094	Significantly worse than best at 1%
	LSTM	LSTM-TS	92	0.4867	1.6879	1.2580	21	Hybrid_EGARCH_TSB	1.5986	0.1134	Not significantly different from best
	Hybrid	Hybrid_GJR_GARCH_TSB	92	0.5138	1.7383	1.1977	22	Hybrid_EGARCH_TSB	2.1075	0.0378	Significantly worse than best at 5%
	LSTM	LSTM-TSB	92	0.5610	1.6634	1.1794	23	Hybrid_EGARCH_TSB	1.6269	0.1072	Not significantly different from best
	Hybrid	Hybrid_iGARCH_T	92	0.6173	1.8558	1.3913	24	Hybrid_EGARCH_TSB	2.3581	0.0205	Significantly worse than best at 5%
	Hybrid	Hybrid_EGARCH_T	92	0.6585	1.8247	1.3798	25	Hybrid_EGARCH_TSB	2.0233	0.0460	Significantly worse than best at 5%
	Hybrid	Hybrid_GJR_GARCH_T	92	0.6984	1.8201	1.2362	26	Hybrid_EGARCH_TSB	2.2053	0.0299	Significantly worse than best at 5%
	Hybrid	Hybrid_TGARCH_T	92	1.0928	2.0258	1.4450	27	Hybrid_EGARCH_TSB	2.1554	0.0338	Significantly worse than best at 5%
Crypto Winter	Hybrid	Hybrid_APARCH_TS	244	0.2638	1.0951	0.9131	1	Hybrid_APARCH_TS			Best benchmark
	Hybrid	Hybrid_GARCH_TS	244	0.2667	1.1805	0.9494	2	Hybrid_APARCH_TS	0.0646	0.9486	Not significantly different from best
	Hybrid	Hybrid_TGARCH_TSB	244	0.2732	1.1909	0.9491	3	Hybrid_APARCH_TS	0.1909	0.8488	Not significantly different from best
	Hybrid	Hybrid_APARCH_TSB	244	0.2752	1.1653	0.9781	4	Hybrid_APARCH_TS	0.2626	0.7931	Not significantly different from best
	Hybrid	Hybrid_TGARCH_TS	244	0.2832	1.2355	1.0272	5	Hybrid_APARCH_TS	0.4574	0.6478	Not significantly different from best
	Hybrid	Hybrid_GJR_GARCH_TSB	244	0.2881	1.2585	1.0117	6	Hybrid_APARCH_TS	0.5580	0.5773	Not significantly different from best
	Hybrid	Hybrid_EGARCH_TS	244	0.2954	1.1946	0.9817	7	Hybrid_APARCH_TS	0.8399	0.4018	Not significantly different from best
	LSTM	LSTM-TS	244	0.2978	1.2470	1.0744	8	Hybrid_APARCH_TS	1.0903	0.2767	Not significantly different from best
	Hybrid	Hybrid_iGARCH_TSB	244	0.3027	1.1913	0.9964	9	Hybrid_APARCH_TS	1.1579	0.2480	Not significantly different from best
	Hybrid	Hybrid_APARCH_T	244	0.3062	1.2224	1.0073	10	Hybrid_APARCH_TS	1.2200	0.2237	Not significantly different from best
	Hybrid	Hybrid_GJR_GARCH_T	244	0.3137	1.2758	1.0628	11	Hybrid_APARCH_TS	1.3527	0.1774	Not significantly different from best
	Hybrid	Hybrid_GARCH_TSB	244	0.3233	1.2998	1.1033	12	Hybrid_APARCH_TS	1.3029	0.1938	Not significantly different from best
	LSTM	LSTM-T	244	0.3286	1.3711	1.1317	13	Hybrid_APARCH_TS	1.2889	0.1986	Not significantly different from best
	LSTM	LSTM-TSB	244	0.3370	1.3346	1.1147	14	Hybrid_APARCH_TS	2.0714	0.0394	Significantly worse than best at 5%
	Hybrid	Hybrid_iGARCH_TS	244	0.3391	1.3422	1.1309	15	Hybrid_APARCH_TS	1.4879	0.1381	Not significantly different from best
	Hybrid	Hybrid_GJR_GARCH_TS	244	0.3422	1.3784	1.1292	16	Hybrid_APARCH_TS	2.0492	0.0415	Significantly worse than best at 5%
	Hybrid	Hybrid_iGARCH_T	244	0.3477	1.4370	1.1871	17	Hybrid_APARCH_TS	1.6223	0.1060	Not significantly different from best
	Hybrid	Hybrid_TGARCH_T	244	0.3602	1.2956	1.0639	18	Hybrid_APARCH_TS	2.1118	0.0357	Significantly worse than best at 5%

	Hybrid	Hybrid_GARCH_T	244	0.3653	1.4490	1.2515	19	Hybrid_APARCH_TS	2.7264	0.0069	Significantly worse than best at 1%
	Hybrid	Hybrid_EGARCH_TSB	244	0.3715	1.4561	1.1656	20	Hybrid_APARCH_TS	2.1078	0.0361	Significantly worse than best at 5%
	Hybrid	Hybrid_EGARCH_T	244	0.3721	1.4771	1.1942	21	Hybrid_APARCH_TS	1.8493	0.0656	Marginally worse than best at 10%
	Traditional	GARCH	244	0.4686	1.6126	1.4044	22	Hybrid_APARCH_TS	3.8856	0.0001	Significantly worse than best at 1%
	Traditional	TGARCH	244	0.4905	1.6539	1.4151	23	Hybrid_APARCH_TS	3.9699	0.0001	Significantly worse than best at 1%
	Traditional	GJR-GARCH	244	0.4908	1.6904	1.4539	24	Hybrid_APARCH_TS	4.0388	0.0001	Significantly worse than best at 1%
	Traditional	iGARCH	244	0.5356	1.7305	1.5066	25	Hybrid_APARCH_TS	3.8109	0.0002	Significantly worse than best at 1%
	Traditional	APARCH	244	1.5810	1.6491E+06	1.1301E+05	26	Hybrid_APARCH_TS	2.8646	0.0045	Significantly worse than best at 1%
	Traditional	EGARCH	244	6.4517E+07	6.5220	2.1463	27	Hybrid_APARCH_TS	1.0058	0.3155	Not significantly different from best

Notes: *For each crisis regime, models are ranked by QLIKE. The regime-best model is the specification with the lowest QLIKE within that crisis window. The DM test compares each model with the corresponding regime-best benchmark using QLIKE loss differentials. A p-value above 0.10 indicates that the model is not statistically worse than the regime-best benchmark. RMSE and MAE are reported only as complementary diagnostics. Non-rejection by the DM test should not be interpreted as practical superiority when the model exhibits extreme QLIKE or scale-based errors.*

The full crisis rankings confirm the main manuscript's regime-dependent conclusion. A hybrid model achieves the lowest QLIKE in each crisis, but the preferred hybrid specification changes across episodes: Hybrid_TGARCH_TS during COVID-19, Hybrid_EGARCH_TSB during the China mining ban, and Hybrid_APARCH_TS during the crypto winter. This supports the interpretation that hybrid gains are not universal, but depend on the interaction between the volatility state, the information set, and the nature of the crisis.

Table S4 summarizes the models that are statistically indistinguishable from the relevant benchmark within each period or regime, grouped by model family. This table is derived from the full rankings in Tables S2 and S3 and helps show whether statistical competitiveness is concentrated within one model class or distributed across traditional, LSTM, and hybrid specifications.

Table S4. Summary of Models Statistically Indistinguishable from the Benchmark by Regime and Model Family

Period/Regime	Model family	Number of competitive models	Best competitive model in family	Best competitive QLIKE in family	Competitive models in family
Normal	Traditional	4	TGARCH	0.2209	TGARCH; GARCH; GJR-GARCH; EGARCH
	LSTM	2	LSTM-T	0.1934	LSTM-T; LSTM-TS
	Hybrid	13	Hybrid_GARCH_T	0.2062	Hybrid_GARCH_T; Hybrid_APARCH_T; Hybrid_iGARCH_T; Hybrid_EGARCH_T; Hybrid_EGARCH_TSB; Hybrid_TGARCH_TS; Hybrid_GARCH_TS; Hybrid_GJR_GARCH_T; Hybrid_iGARCH_TS; Hybrid_APARCH_TS; Hybrid_EGARCH_TS; Hybrid_GJR_GARCH_TSB; Hybrid_GJR_GARCH_TS
COVID-19 Shock	Traditional	0			
	LSTM	1	LSTM-TSB	2.2881	LSTM-TSB
	Hybrid	10	Hybrid_TGARCH_TS	1.2636	Hybrid_TGARCH_TS; Hybrid_GARCH_TSB; Hybrid_GJR_GARCH_TS; Hybrid_EGARCH_TS; Hybrid_APARCH_TS; Hybrid_GJR_GARCH_TSB; Hybrid_iGARCH_TSB; Hybrid_GARCH_TS; Hybrid_TGARCH_TSB; Hybrid_EGARCH_TSB
China Mining Ban	Traditional	4	GARCH	0.1870	GARCH; EGARCH; iGARCH; TGARCH
	LSTM	3	LSTM-T	0.4074	LSTM-T; LSTM-TS; LSTM-TSB
	Hybrid	7	Hybrid_EGARCH_TSB	0.1528	Hybrid_EGARCH_TSB; Hybrid_EGARCH_TS; Hybrid_GJR_GARCH_TS; Hybrid_APARCH_TS; Hybrid_TGARCH_TS; Hybrid_iGARCH_TS; Hybrid_GARCH_TS
Crypto Winter	Traditional	1	EGARCH	6.4517E+07	EGARCH
	LSTM	2	LSTM-TS	0.2978	LSTM-TS; LSTM-T
	Hybrid	13	Hybrid_APARCH_TS	0.2638	Hybrid_APARCH_TS; Hybrid_GARCH_TS; Hybrid_TGARCH_TSB; Hybrid_APARCH_TSB; Hybrid_TGARCH_TS; Hybrid_GJR_GARCH_TSB; Hybrid_EGARCH_TS; Hybrid_iGARCH_TSB; Hybrid_APARCH_T; Hybrid_GJR_GARCH_T; Hybrid_GARCH_TSB; Hybrid_iGARCH_TS; Hybrid_iGARCH_T

Notes: *A model is classified as statistically indistinguishable from the benchmark if it is either the period/regime-best model or if the DM-test p-value relative to the benchmark exceeds 0.10. This classification reflects statistical non-rejection of equal predictive accuracy, not necessarily practical model preference. Models with extreme QLIKE or scale-based errors should therefore be interpreted with caution.*

Table S5 provides a compact regime-level summary of the statistically indistinguishable model set. It reports the benchmark model in each period or crisis regime, the benchmark family, the benchmark QLIKE, and the number of traditional, LSTM, and hybrid models that are not statistically worse than the benchmark according to the Diebold–Mariano test.

Table S5. Regime-Level Summary of Models Statistically Indistinguishable from the Benchmark

Period/Regime	Observations	Regime-best model	Regime-best family	Regime-best QLIKE	Number of competitive models	Competitive traditional models	Competitive LSTM models	Competitive hybrid models	Competitive criterion	Competitive models
Normal	1583	LSTM-T	LSTM	0.1934	19	4	2	13	Regime-best OR p-value vs best > 0.10	LSTM-T; Hybrid_GARCH_T; Hybrid_APARCH_T; Hybrid_iGARCH_T; Hybrid_EGARCH_T; Hybrid_EGARCH_TSB; Hybrid_TGARCH_TS; Hybrid_GARCH_TS; TGARCH; GARCH; Hybrid_GJR_GARCH_T; Hybrid_iGARCH_TS; Hybrid_APARCH_TS; GJR-GARCH; LSTM-TS; Hybrid_EGARCH_TS; Hybrid_GJR_GARCH_TSB; Hybrid_GJR_GARCH_TS; EGARCH
COVID-19 Shock	102	Hybrid_TGARCH_TS	Hybrid	1.2636	11	0	1	10	Regime-best OR p-value vs best > 0.10	Hybrid_TGARCH_TS; Hybrid_GARCH_TSB; Hybrid_GJR_GARCH_TS; Hybrid_EGARCH_TS; Hybrid_APARCH_TS; Hybrid_GJR_GARCH_TSB; Hybrid_iGARCH_TSB; Hybrid_GARCH_TS; LSTM-TSB; Hybrid_TGARCH_TSB; Hybrid_EGARCH_TSB
China Mining Ban	92	Hybrid_EGARCH_TSB	Hybrid	0.1528	14	4	3	7	Regime-best OR p-value vs best > 0.10	Hybrid_EGARCH_TSB; GARCH; EGARCH; iGARCH; TGARCH; Hybrid_EGARCH_TS; Hybrid_GJR_GARCH_TS; Hybrid_APARCH_TS; Hybrid_TGARCH_TS; Hybrid_iGARCH_TS; Hybrid_GARCH_TS; LSTM-T; LSTM-TS; LSTM-TSB
Crypto Winter	244	Hybrid_APARCH_TS	Hybrid	0.2638	16	1	2	13	Regime-best OR p-value vs best > 0.10	Hybrid_APARCH_TS; Hybrid_GARCH_TS; Hybrid_TGARCH_TSB; Hybrid_APARCH_TSB; Hybrid_TGARCH_TS; Hybrid_GJR_GARCH_TSB; Hybrid_EGARCH_TS; LSTM-TS; Hybrid_iGARCH_TSB; Hybrid_APARCH_T; Hybrid_GJR_GARCH_T; Hybrid_GARCH_TSB; LSTM-T; Hybrid_iGARCH_TS; Hybrid_iGARCH_T; EGARCH

Notes: *The competitive criterion is defined as being the period/regime-best model or having a DM-test p-value above 0.10 relative to the benchmark. The table should be read together with Tables S2 and S3, because statistical non-rejection does not override the QLIKE ranking or evidence of numerical instability.*

Overall, the supplementary rankings and summary tables reinforce the main manuscript's conclusion that Bitcoin volatility forecast performance is regime-dependent. In normal periods, several traditional, LSTM, and hybrid models are statistically close to the LSTM-T benchmark. During crises, the lowest QLIKE is always achieved by a hybrid model, but the relevant hybrid specification differs across crisis episodes. The supplementary evidence therefore supports a regime-sensitive interpretation rather than an unconditional claim of model-class superiority.